

Lecture Notes on Differential Equations and Transform Techniques

Lecture Notes for Spring 2025

The purpose of these lecture notes is to provide a concise summary of the key concepts covered in the course. They are designed to serve as a **quick reference** and should be used in conjunction with the core textbooks listed below. While some sections are adapted from these referenced texts, the notes are not intended to replace the original works, which offer more comprehensive explanations. To gain a deeper understanding, students are encouraged to refer to the textbooks listed.

References:

1. W. E. Boyce, R. C. DiPrima, D. B. Meade, *Elementary Differential Equations and Boundary Value Problems*, 9th edition, Wiley, 1997.
2. Brad G. Osgood, *Lectures on Fourier Transform and Its Applications*, IAS/Park City Mathematics Series / American Mathematical Society (AMS), 2019.
3. Charles Henry Edwards, David E. Penney, *Differential Equations and Boundary Value Problems: Computing and Modeling*, 4th edition, Pearson, 2000.
4. George F. Simmons, *Differential Equations with Applications and Historical Notes*, 2nd edition, CRC Press, 2016.
5. Ruel V. Churchill, *Fourier Series and Boundary Value Problems*, McGraw-Hill, 1941.
6. Gustav Doetsch, *Introduction to the Theory and Application of the Laplace Transformation*, Springer, 2012.
7. Morris Tenenbaum, Harry Pollard, *Ordinary Differential Equations*, Dover Books on Mathematics, Dover Publications, 1985.

EE1060: Differential Equations and Transform Techniques

Lecture Notes - 05

Topics covered :

1. Integration Factor Approach

- Consider the first-order differential equation given by

$$y \, dx + nx \, dy = 0. \quad (1)$$

This equation is non-exact. Although it is not immediately solvable using the standard methods for exact equations, we can transform this non-exact equation into an exact one by multiplying through by an appropriate function. In this case, multiplying the equation by y^{n-1} makes it exact.

- Another Example:** Consider the following differential equation [1]

$$y(x^4y - 1)dx + x(y^4x - 1)dy = 0 \quad (2)$$

This equation is not exact, since

$$\frac{\partial P}{\partial y} = (2yx^4 - 1) \text{ and } \frac{\partial Q}{\partial x} = (2xy^4 - 1) \quad (3)$$

are not equal. However, by multiplying the equation by the integrating factor $\frac{1}{x^2y^2}$, we transform the equation into an exact one. Multiplying the given ODE with the integrating factor gives

$$\underbrace{\left(x^2 - \frac{1}{x^2y}\right)}_M dx + \underbrace{\left(y^2 - \frac{1}{y^2x}\right)}_N dy = 0 \quad (4)$$

Equation (4) is exact since

$$\frac{\partial M}{\partial y} = \frac{-1}{x^2y^2} \text{ and } \frac{\partial N}{\partial x} = -\frac{1}{x^2y^2} \quad (5)$$

are equal.

- The idea of multiplying a non-exact differential equation by a suitable function to make it exact is the core idea behind the method of integrating factors. This approach relies on the observation that if a differential equation is not easily solvable in its original form, we can often find a multiplicative factor that transforms it into a solvable one. However, it's important to note that identifying the correct integrating factor is not always straightforward. While many types of differential equations can be made exact with the right factor, in this course we will focus on a select few.
- Formally, an integrating factor is defined as a multiplicative factor that transforms an inexact differential equation into an exact one [1]. However, in the context of this course, particularly for first-order ODEs, we define an

integrating factor as a multiplicative factor that converts a differential equation, which is not easily solvable in its original form, into a form that is more straightforward to solve.

- **Integrating factor for non-exact first-order ODE:** Consider a non-exact first-order ODE

$$P(x, y)dx + Q(x, y)dy \quad (6)$$

A factor $u(x, y)$ constitutes an integrating factor if

$$\begin{aligned} \frac{\partial u(x, y)P(x, y)}{\partial y} &= \frac{\partial u(x, y)Q(x, y)}{\partial x} \\ \text{i.e., } P(x, y)\frac{\partial u(x, y)}{\partial y} + u(x, y)\frac{\partial P(x, y)}{\partial y} &= Q(x, y)\frac{\partial u(x, y)}{\partial x} + u(x, y)\frac{\partial Q(x, y)}{\partial x} \\ P(x, y)\frac{\partial u(x, y)}{\partial y} - Q(x, y)\frac{\partial u(x, y)}{\partial x} &= u(x, y)\left[\frac{\partial Q(x, y)}{\partial x} - \frac{\partial P(x, y)}{\partial y}\right] \end{aligned} \quad (7)$$

Equation (7) indicates that finding $u(x, y)$ in general requires solving a partial differential equation. However, in certain cases, a simple choice of $u(x, y)$ can convert a non-exact first-order ODE into an exact one. We will explore two such scenarios.

1. For the integrating factor to be purely a function of x i.e., $u(x, y) = u(x)$, the term on the right-hand side of (8)

$$\frac{1}{u(x)} \frac{\partial u(x)}{\partial x} = \frac{1}{Q(x, y)} \left[\frac{\partial P(x, y)}{\partial y} - \frac{\partial Q(x, y)}{\partial x} \right] \quad (\text{from (7) by setting } u(x, y) = u(x) \implies \frac{\partial u}{\partial y} = 0) \quad (8)$$

must be a function of x alone. In such a scenario, $u(x)$ can be computed by integrating (8) on both sides i.e.,

$$\frac{\partial u}{u} = h(x)dx \implies u(x) = \pm e^{\int h(x)dx} \quad (9)$$

$$\text{where } h(x) = \frac{1}{Q(x, y)} \left[\frac{\partial P(x, y)}{\partial y} - \frac{\partial Q(x, y)}{\partial x} \right].$$

2. For the integrating factor to be purely a function of y i.e., $u(x, y) = u(y)$, the term on the right-hand side of (10)

$$\frac{1}{u(y)} \frac{\partial u(y)}{\partial y} = \frac{1}{P(x, y)} \left[\frac{\partial Q(x, y)}{\partial x} - \frac{\partial P(x, y)}{\partial y} \right] \quad (\text{from (7) by setting } u(x, y) = u(y) \implies \frac{\partial u}{\partial x} = 0) \quad (10)$$

must be a function of y alone. In such a scenario, $u(y)$ can be computed by integrating (10) on both sides i.e.,

$$\frac{\partial u}{u} = h(y)dy \implies u(y) = \pm e^{\int h(y)dy} \quad (11)$$

$$\text{where } h(y) = \frac{1}{P(x, y)} \left[\frac{\partial Q(x, y)}{\partial x} - \frac{\partial P(x, y)}{\partial y} \right].$$

- **Integrating factor for first-order linear ODE [2]:** Consider the first-order linear ODE

$$\frac{dy}{dx} + p(x)y = q(x). \quad (12)$$

We aim to find an integrating factor $u(x)$ such that multiplying the equation by $u(x)$ transforms it into a form that can be easily solved. Specifically, we want the resulting equation to become separable. Multiplying 12 by $u(x)$ gives

$$u(x) \frac{dy}{dx} + u(x)p(x)y = q(x)u(x). \quad (13)$$

To make the (13) separable, the left-hand side of the equation can be expressed as the derivative of a product $u(x)y(x)$ if $u(x)$ is such that

$$u'(x) = u(x)p(x) \implies \frac{1}{u(x)} \frac{du}{dx} = p(x) \quad (14)$$

One choice of $u(x)$ can be obtained by integrating (14) on both sides, which gives

$$u(x) = e^{\int p(x) dx} \quad (15)$$

Thus, the integrating factor for the first-order linear ODE of the form (12) is $u(x) = e^{\int p(x) dx}$. For this choice of integrating factor, the solution of (12) can be computed as

$$\frac{d}{dx}(u(x)y) = u(x)q(x) \quad (16)$$

Integrating 16 on both sides gives

$$u(x)y = \int u(x)q(x) + c \quad (17)$$

References

- [1] M. Tenenbaum and H. Pollard, *Ordinary Differential Equations*, reprint of the 1963 original ed. Dover Publications, 1985, dover Books on Mathematics.
- [2] C. H. Edwards and D. E. Penney, *Differential Equations and Boundary Value Problems: Computing and Modeling*, 4th ed. Upper Saddle River, NJ: Pearson, 2000.